# CE205 Databases and Information Retrieval

## Class 4

### Feature Selection using Fisher Discriminant Ratio (FDR)

#### How to calculate the fisher discriminant ratio

**Step 1:** Calculate the means of the corresponding features of both classes.

**Step 2:** Calculate the standard deviations of the corresponding features of both classes

**Step 3:** Use the following formula

$$FDR = \frac{\left[mean(C1) - mean(C2)\right]^2}{\left[std(C1)\right]^2 + \left[std(C2)\right]^2}$$

where

***C1*** is class1 and ***C2*** is class2

***Std*** represents the standard deviation

**Guideline to the solution:**

**Class 1** **Class2**

| Feature | 1 | 2 | 3 | 4 | Feature | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|---|---|---|---|
| *Mean* (C1) | 19.4 | 12.57 | 3.04 | 6.79 | *Mean* (C2) | 39.6 | 23.02 | 6.56 | 16.59 |
| *Std* (C1) | 12.9 | 7.14 | 3.06 | 7.49 | *Std* (C2) | 17.7 | 7.80 | 3.89 | 13.87 |

Can you conclude which features are more discriminative for the classification of the given classes?

## How to predict the class of samples when models are given

| Model for class 1 (**setosa**) contains three means | Model for class 2 (**versicolor**) contains three means |
|---|---|
| CS1 (1.42 , 0.22 ) | VC1 (4.81 , 1.54) |
| CS2 (1.64 , 0.31 ) | VC2 ( 3.78 , 1.14) |
| CS3 (1.10 , 0.17) | VC3 ( 4.48 , 1.41) |

Predict class for the following test samples

( 1.20 , 0.20 )

(4.60 , 1.20)

(1.40 , 0.10)

**Guideline to the solution:**

- First, compute the distance of a test sample with all means of **setosa** (*follow slide 28 of Lec4b*)
- Then, compute the average distance (*follow slide 29 of Lec4b*)
- Now, compute the distance of a test sample with all means of **versicolor** (*follow slide 28 of Lec4b*)
- Then, compute the average distance (*follow slide 29 of Lec4b*)
- Make your decision: